05 NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

62.1 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



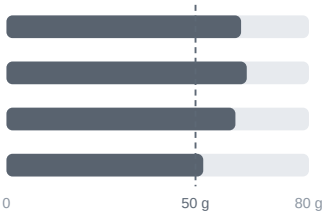
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 16.0 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**39.1 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 2.2 counterbore**
Material volume / surface	**50.04 cm³ / 446 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**62.1 g**
PETG	1.27 g/cm³	**63.6 g**
TPU	1.21 g/cm³	**60.6 g**
ABS	1.04 g/cm³	**52.0 g**



COLOURWAY

#18181A #CE1820

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**22.3 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 20.8 m (100 % flow)**

DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

Ridges and bumps are gentle slopes; the network is printed in the same material as the base.

KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

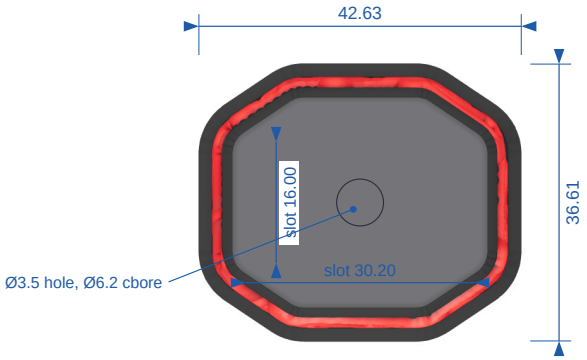
VERIFIED ON THE STL

Paddle slot	**30.2 × 16.0 mm (30 + 0.2, 16 space)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 2.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 7**
Wall around the cavity (min)	**1.10 mm**
Material under 0.4 mm wide	**0.000 % of volume**
Widest flat bridge	**3.4 mm**

FILES

stl/05_neuro_tread.stl	23.5 MB · 469,618 triangles
3mf/05_neuro_tread.3mf	5.0 MB · colour

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

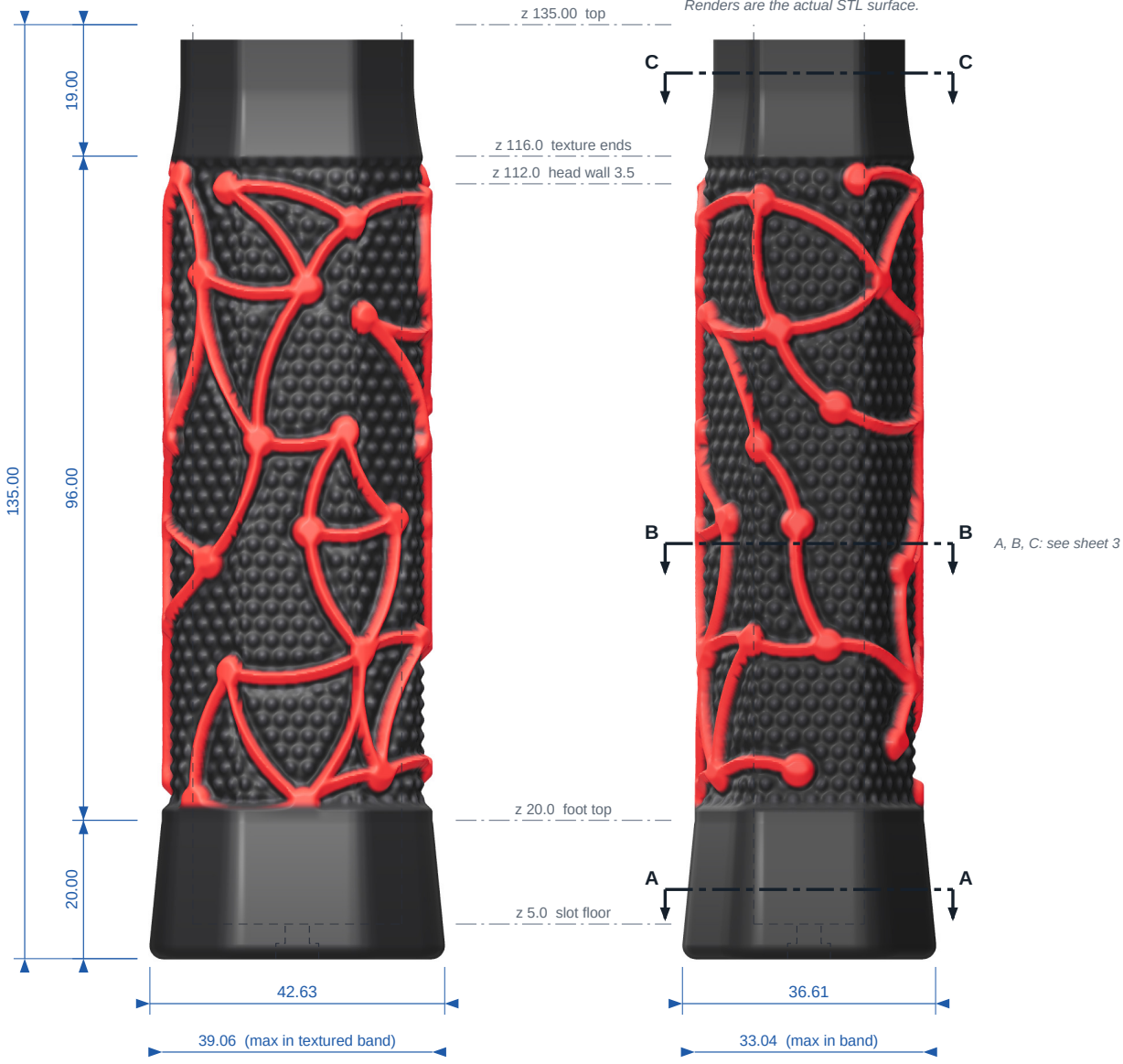


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 16.00**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**39.06 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 2.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1

0 10 20 30 40 50 mm

PART

NEURO-TREAD

Pickleball paddle grip sleeve, closed butt, open throat

SCALE	UNITS	PROJECTION
1 : 1	millimetres	
MATERIAL	**INFILL**	**LAYER HEIGHT**
PLA / PETG / TPU	100 %	0.20 mm
MASS (PLA)	**VOLUME**	**SURFACE AREA**
62.1 g	50.04 cm³	446 cm²
REVISION	**DATE**	**SHEET**
8	2026-10-23	2 of 4

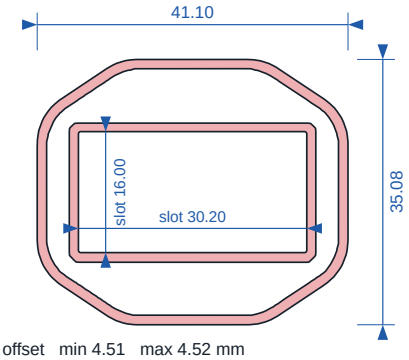
GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 100 solid, end walls slit 20 mm.
4. Outer envelope matches PickleballGrip_1.3mf below z 90; head thickened.

Sections & profiles

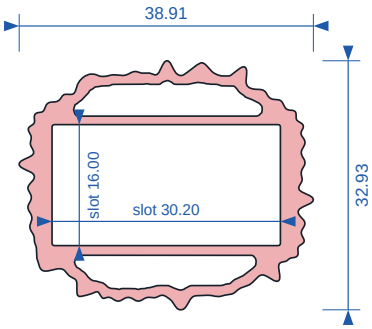
True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

SECTION A–A — z = 10 mm · foot



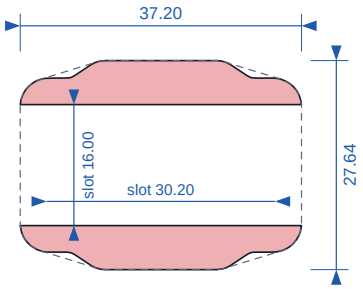
Skin offset min 4.51 max 4.52 mm
Area 277 mm² · outer perimeter 128.9 mm

SECTION B–B — z = 60 mm · textured band



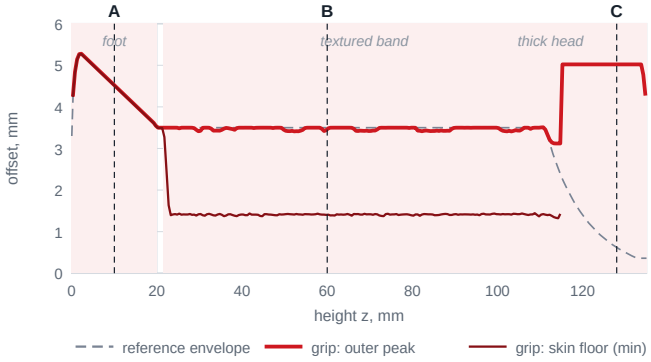
Skin offset min 1.41 max 3.46 mm
Area 303 mm² · outer perimeter 116.9 mm

SECTION C–C — z = 128 mm · rim

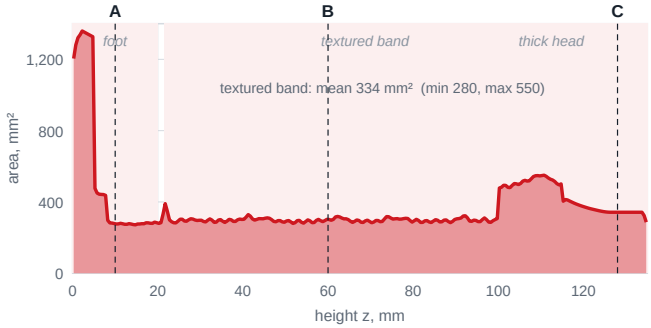


Skin offset max 5.02 mm
Area 342 mm² · outer perimeter 115.3 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 16.00 mm**
Slot area / perimeter	**483.1 mm² / 91.9 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 2.2**
Sealed cavity / min wall	**22.3 cm³ / 1.10 mm**
Outer W × D in band (max)	**39.06 × 33.04 mm**
Material volume / surface	**50.04 cm³ / 445.6 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck	116.0	112.0	**-4.0**
Head (solid, thick wall, slot opening)	112.0	135.00	**23.0**

Relief floor above reference line (design / min)	**1.20 / 1.37 mm**
Slot wall over the head (design / min)	**3.50 / 3.50 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

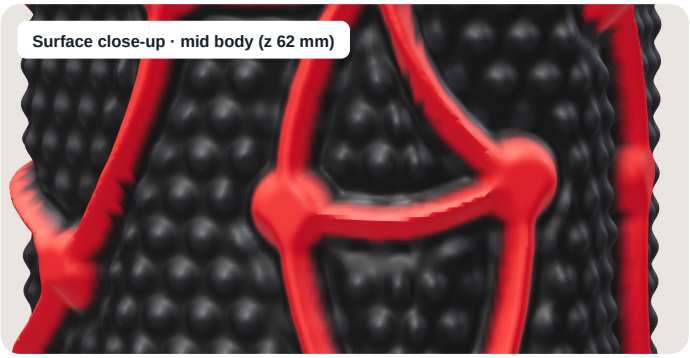
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	902	5.00	5.01
10.0	41.10	35.08	128.9	278	4.51	4.52
20.0	39.11	33.09	122.6	282	3.52	3.52
30.0	37.77	31.75	116.8	297	1.42	3.42
45.0	38.99	32.88	117.1	307	1.40	3.50
60.0	38.91	32.93	116.9	302	1.41	3.46
75.0	38.94	32.96	119.3	307	1.41	3.50
90.0	38.99	32.86	117.2	303	1.42	3.50
105.0	38.81	32.79	117.7	501	1.42	3.43
116.0	37.20	30.19	117.3	412	open	5.02
125.0	37.20	27.75	115.4	345	open	5.02
132.0	37.20	27.64	115.3	342	open	5.02

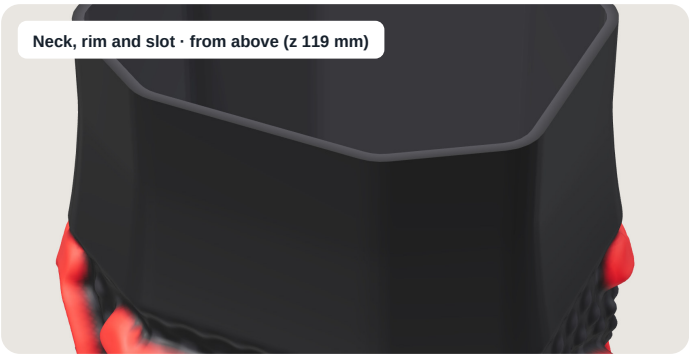
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

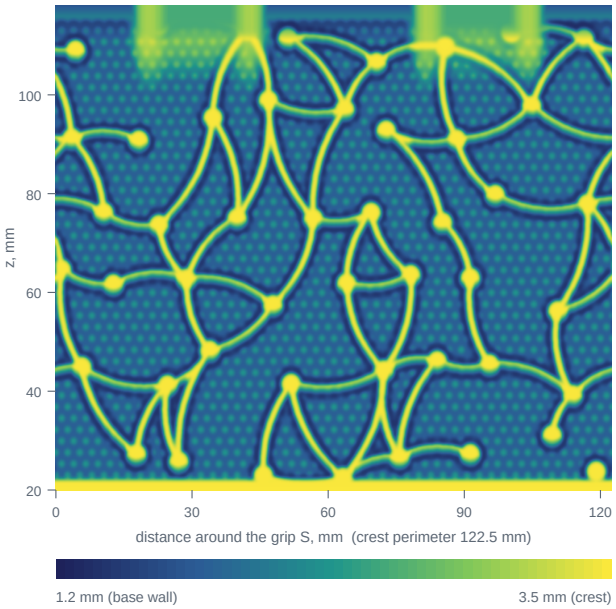


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Neurons (somas)	**48 (8 × 6)**
Axon ridge width	**about 1.5 mm**
Soma diameter	**about 4.2 mm**
Network height	**3.50 mm above base**
Bump pitch	**2.66 mm (hex)**
Bump field height	**1.4 – 2.4 mm above base**
Ridge / bump colours	**red / black**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 7**
Wall around the cavity (min)	**1.10 mm**
Paddle slot	**30.2 × 16.0 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.000 %**
Widest flat bridge	**3.4 mm**